Project Interim Report

On

File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming With Java

SESSION: 2025-26

Submitted By

Student name

Reg No.

Bachelor of Technology in Computer Science & Engineering with AI and ML in association with IBM

Under the Guidance of

Mr. Santanu Sasmal

IBM SME and EXPERT

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

RUNGTA INTERNATIONAL SKILLS UNIVERSITY

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Abstract

Short summary about your project in 150-200 words. Include the following

· What is your project

· Why is it needed

· What it does

Introduction

Same information about your project but in details you can also add some of the features of your project

Objectives (what your project does)

- Manage book records
- Track issued books
- Reduce manual work
- Apply OOP concepts

Scope

This project is suitable for small libraries and can be extended with database integration in future.

System Requirements

Hardware: Computer with 4GB RAM

Software: Java JDK, IDE (VS Code/Eclipse/Edit Plus)

Methodology

Explain how your project works like first we add something, then search, then modify etc.

Flowchart

Includes Flowchart (Start, Menu, Choice, Operations) , and Class Diagram representing system structure.

Sample class diagram –

· **ClassName**

o Variables inside class

o Methods inside class

o If array is present within the class also mention it here

Implementation

Explain different classes used in your program(only explain logic don't paste code here)

Sample Input/Output

Paste screenshot of your terminal with different operations

Future Enhancements

Things you can improve on your project

Implementation

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Sample Input/Output
Paste screenshot of your terminal with different operations
Future Enhancements
Things you can improve on your project

Gantt Chart

1. Planning & Structure

A Gantt chart helps you break the project into tasks (design, coding, testing, etc.) and set timelines. Even if it's already built once, rebuilding or modifying still needs planning.

2. Time Management

It shows:

- When each task starts
- When it should finish
- This helps avoid delays and last-minute panic.

Conclusion

The project successfully demonstrates Java OOP concepts and provides a functional system.

References

Book

Format:

[1] Author(s), *Book Title*, Edition. City: Publisher, Year.

Example:

[1] H. Schildt, *Java: The Complete Reference*, 11th ed. New York: McGraw-Hill, 2018.

Website

Format:

[3] Author (if any), "Title of page," Website Name. [Online]. Available: URL

Example:

[3] Oracle, "Java Documentation," Oracle. [Online]. Available: https://docs.oracle.com

Conference Paper

Format:

[4] Author(s), "Paper title," in Proc. Conference Name, year, pp.

Example:

[4] S. Kumar, "AI in Education," in Proc. IEEE Conf. on AI, 2021, pp. 120-125.

Journal Article

Format:

[2] Author(s), "Title of paper," Journal Name, vol., no., pp., year.

Example:

[2] A. Sharma and R. Gupta, "Machine Learning Techniques," International Journal of AI, vol. 5, no. 2, pp. 45-50, 2020.